入侵澳洲的毒蟾

The Toad That Invaded Australia



引进的灭虫帮手，为什么成了生态难题？

READ 纯净阅读

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RECALL 回想巩固

英文原文

Greyback cane beetles, whose larvae devour roots, were plaguing lucrative sugar plantations in Queensland, north-eastern Australia. In June 1935 Queensland imported 102 cane toads, bulky amphibians native to the Americas, from Hawaii. That August about 2,400 of their young were released. Biological control, setting one species against a pest, had an ambitious new recruit.

The hasty plan met opposition. In November 1935 an insect expert, Walter Froggatt, persuaded officials to halt releases; after Queensland's lobbying, they resumed in September 1936. The scheme flopped: the toads barely dented beetle numbers, perhaps because the insects stayed high up on cane stalks.

The toads thrived instead. With few foes and fertile females spawning thousands of eggs, they multiplied unchecked. Their invasion frontier, creeping about ten kilometres a year at first, accelerated to about 50 a year in the tropical north. In February 2009 the first known toad breached Western Australia's border.

Toxin glands in their skin make them a deadly meal. Numbers of goannas (large lizards) and northern quolls (small marsupial hunters) have slumped in places; some local quoll populations were wiped out. The toads have wreaked havoc, but the damage is uneven; some species have prospered.

Biological control can triumph. From 1926 the cactus moth, Cactoblastis, cleared most prickly pear, an invasive cactus, from Queensland and New South Wales within about a decade. Tested first, its larvae starved rather than eat fifteen other plants; the toads faced no such test. Under an October 1984 law, recruits now face stringent scrutiny. The blunder was born of haste and sealed by lobbying that overrode a warning: an untested ally, once loose, may never be recalled.

中文译文

在澳大利亚东北部昆士兰州，灰背蔗甲虫的幼虫啃食根系，困扰着利润丰厚的甘蔗种植业。1935年6月，当地从夏威夷引进102只甘蔗蟾蜍。这些体型笨重的两栖动物原产美洲。同年8月，约2400只繁育出的幼蟾被放到野外。用一种生物控制另一种害虫的生物防治，就这样迎来了一位雄心勃勃的“新兵”。

这项草率的计划遭到反对。1935年11月，昆虫专家沃尔特·弗罗格特说服官员暂停释放，但昆士兰州游说后，释放又在1936年9月恢复。计划最终失败：蟾蜍几乎没有减少甲虫数量，可能因为这些昆虫待在高高的甘蔗茎秆上。

蟾蜍自身反而繁盛起来。它们天敌很少，雌蟾能产下数千枚卵，于是数量不受控制地增加。入侵前沿最初每年推进约10公里，在热带北部加速到每年约50公里。2009年2月，已知的第一只蟾蜍越过了西澳大利亚州边界。

蟾蜍皮肤里有毒腺，捕食者吃下它们可能送命。一些地区的巨蜥和北袋鼬数量大幅下降，某些地方的袋鼬种群甚至消失了。巨蜥是大型蜥蜴，北袋鼬则是小型有袋类捕食者。蟾蜍造成了严重生态破坏，但影响并不均匀，一些物种反而兴盛起来。

生物防治也有成功的例子。自1926年起，仙人掌螟在约十年间清除了昆士兰州和新南威尔士州大部分入侵的仙人掌属植物。它的幼虫事先经过检验，宁可饿死也不吃另外15种植物；蟾蜍却没有接受过这样的测试。根据1984年10月通过的法律，如今引入生物防治物种需要严格审查。这次失误始于仓促，又因游说压过警告而成为定局：一个未经检验的“盟友”，一旦放出，就可能再也召不回来。

原文只说甲虫在高处“可能”解释防治失败；生态损害也并非对所有物种相同。

核心词汇

先记这22项。连着下方的原文短语一起回想。

larva n.
幼虫
whose larvae devour roots

devour v.
大量啃食
larvae devour roots

lucrative adj.
利润丰厚的
lucrative sugar plantations

amphibian n.
两栖动物
bulky amphibians

native to 形容词短语
原产于
native to the Americas

species n.
物种
setting one species against a pest

pest n.
害虫；有害生物
one species against a pest

hasty adj.
草率仓促的
The hasty plan met opposition

opposition n.
反对
The hasty plan met opposition

persuade v.
说服
persuaded officials to halt releases

lobbying n.
游说
after Queensland's lobbying

resume v.
恢复；重新开始
they resumed in September 1936

thrive v.
繁盛
The toads thrived instead

unchecked adj.
未受遏制的
they multiplied unchecked

invasion n.
入侵
Their invasion frontier

accelerate v.
加速
accelerated to about 50 a year

toxin n.
毒素
Toxin glands in their skin

wipe out 动词短语
彻底消灭
quoll populations were wiped out

invasive adj.
入侵性的
an invasive cactus

stringent adj.
严格的
stringent scrutiny

scrutiny n.
仔细审查
stringent scrutiny

override v.
压过；推翻
lobbying that overrode a warning

文中词汇与搭配

补充词汇共40项。保留每个表达在本文中的意思。

beetle n.
甲虫

plague v.
困扰

plantation n.
种植园

import v.
引进

bulky adj.
体积庞大笨重的

release v.
释放

biological adj.
生物的

ambitious adj.
雄心勃勃的

recruit n.
新成员；新兵

halt v.
停止

scheme n.
计划

flop v.
彻底失败

dent v.
削减；产生影响

stalk n.
茎秆

foe n.
敌手；此处指天敌

fertile adj.
能繁殖的

spawn v.
产卵

multiply v.
大量繁殖

frontier n.
前沿

tropical adj.
热带的

文中词汇与搭配 续

补充词汇共40项。保留每个表达在本文中的意思。

breach v.
突破；越过

gland n.
腺体

lizard n.
蜥蜴

marsupial adj.
有袋类的

slump v.
锐减

wreak v.
造成（破坏）

havoc n.
严重破坏

uneven adj.
不均等的

prosper v.
兴盛

triumph v.
大获成功

moth n.
蛾

prickly adj.
多刺的

cactus n.
仙人掌

starve v.
饿死

blunder n.
严重错误

haste n.
仓促

untested adj.
未经检验的

ally n.
盟友

loose adj.
被放出的；不受约束的

recall v.
召回

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Greyback cane beetles, whose larvae devour roots, were plaguing _____ sugar plantations in Queensland, north-eastern Australia.

02 The hasty plan met _____.

03 With few foes and fertile females spawning thousands of eggs, they multiplied _____.

04 Under an October 1984 law, recruits now face stringent _____.

文章主线

- 引进 1935年，澳大利亚昆士兰引进102只甘蔗蟾蜍，放出约2,400只后代，想控制啃食甘蔗根系的甲虫。
- 警告 昆虫专家曾促成暂停释放，地方游说却使1936年恢复；蟾蜍释放前缺少食性检验，警告最终被压过。
- 失控 甲虫几乎没减少，或因它们待在高处；蟾蜍却天敌少、产卵多，在热带北部每年扩散约50公里。
- 伤害 蟾蜍的皮肤毒素使巨蜥、北袋鼬在部分地区锐减，某些袋鼬种群消失；也有物种反而兴盛。
- 对照 仙人掌螟经检验不吃另15种植物，后来成功控制入侵仙人掌；1984年立法要求严格审查生物防治引种。

参考答案　1 lucrative 2 opposition 3 unchecked 4 scrutiny